

CICADA 2067 - Website Requirements Specification

1. Project Overview

The CICADA website is **not** intended to function as a conventional quiz platform. It should provide an immersive, story-driven experience where participants progressively uncover mission archives, classified documents, system logs, and transmissions.

The website should resemble a futuristic spacecraft operating system and maintain continuity with the event storyline.

The application should support multiple rounds with secure authentication, progress tracking, answer validation, leaderboard generation, and administrative controls.

2. User Roles

There will be two user roles.

1. Participant

Capabilities:

- Login using Team ID
- Access current round
- Submit answers
- View unlocked story fragments
- View current progress
- View timer
- Access unlocked files
- Play audio/video assets
- View/download provided documents

2. Administrator

Capabilities:

- Create/Edit/Delete Teams
- Reset Team Progress
- Override Challenge Completion
- Unlock Individual Challenges
- Lock/Unlock Entire Rounds
- View Submission Logs
- View and Export Leaderboard
- Manage Story Assets
- Update Answers
- Monitor Live Progress and

- Auto-qualification to next rounds (override by admin)

3. Authentication Module

Team Registration

Fields

- Team Name
- Team Leader (Has privilege to submit answers)
- Email auth
- Member Names
- Generated Team ID

Login

Participants login using

email

Session should persist.

Automatic logout after inactivity (optional).

4. Dashboard

After login participants should land on a Mission Dashboard.

Example

ENDURANCE OPERATING SYSTEM

Mission Day : 213

Crew Status : Active

Recovered Archives : 2 / 6

Current Objective

Continue Investigation

Dashboard Components

- Current Round
- Current Challenge
- Progress Bar
- Story Progress
- Remaining Time
- Logout

5. Challenge Engine

Each challenge should have

Challenge Name

Example

Archive 03

Crew Research Log

Story Context

Small description introducing the archive. Could be text/audio/video files embedded at ebg/end of rounds

Embedded Assets

Support should exist for

- Images
- PDFs
- Audio
- Videos
- Downloadable Files
- Text Files

Answer Box

Accept

- Text
- Numbers
- Alphanumeric
- Symbols (optional)

Case insensitive comparison.

Ignore leading/trailing spaces.

Validation

Correct answer

↓

Unlock next challenge automatically.(Override by admin privilege required)

Incorrect answer

↓

Display

Incorrect Authentication Key

Try Again

No information leakage.

6. Story Fragment System

Every completed challenge unlocks a story fragment.

Displayed as

Recovered Mission Log

Day 172

The anomaly has intensified...

Fragments should remain accessible from an Archive page.

Participants should be able to revisit previously unlocked fragments.

7. Progress Tracking (visible to admin only)

Track

- Current Challenge
- Challenges Solved
- Time Taken
- Number of Attempts
- Timestamp
- Story Progress

Progress saved automatically.

If participant logs out



Continue from previous point.

8. File Handling

Website should support

PDF Viewer

Embedded.

Download option.

Image Viewer

Zoom

Pan

Fullscreen

Audio Player

Requirements

Play

Pause

Replay

Seek

Volume

Playback speed optional

Supports

mp3

wav

ogg

Video Support

Embedded player.

Fullscreen.

Autoplay disabled.

Supports

mp4

webm

9. Terminal Interface

One of the major UI components.

The website should include a terminal-style interface.

Capabilities

- Typing animation
- Command output
- Fake loading bars
- Progress percentage
- System Logs
- Error Messages

Example

ENDURANCE COMMAND TERMINAL

Loading Navigation Module...

Checking Archive Integrity...

Recovered 31%

Access Granted

Should feel like a spaceship operating system.

10. UI Requirements

Theme

Dark

Minimal

Sci-Fi

Blue / Cyan accents

Glassmorphism

Terminal inspired

Animations

Required

- Fade
- Glitch
- Scan lines
- Typing effect
- Flickering lights
- Progress loading

Typography

Monospace for terminal

Readable font for story

Sound Effects (optional)

Terminal beep

Transmission static

Notification sound

Access Granted

Access Denied

11. Leaderboard (visible to admin)

Live Leaderboard

Fields

Rank

Team Name

Challenges Completed

Completion Time

Sorting based on:

1.Challenges Solved

2.Time Taken

12. Timer

Round Timer

Countdown.

Visible throughout.

Auto submit required.

13. Story Engine

Every challenge unlocks

Story Fragment

Story fragments displayed with

Typing animation.

Fade transition.

Glitch effect.

Audio cue optional.

14. Admin Dashboard

Admin Login

Separate authentication.

Team Management

View Teams

Search Teams

Create Team

Delete Team

Reset Password

Edit Details

Challenge Control

Unlock Challenge

Lock Challenge

Skip Challenge

Reset Challenge

Override Completion

Change Answer

Enable Hint

Disable Hint

Progress Override

Required.

If a team qualifies manually.

Admin should be able to

Team

↓

Advance to Round 2

without solving previous challenge.

Similarly

Round 3 qualification.

Submission Logs

Store

Team

Challenge

Submitted Answer

Correct/Incorrect

Timestamp

Attempts

Searchable.

Leaderboard Export

CSV

Excel

PDF optional.

15. Asset Management

Admin should upload

Images

Audio

Videos

PDFs

Text Files

without touching code.

16. Security

- Session based authentication
- Prevent URL manipulation
- Prevent direct challenge access
- Disable browser caching for answers
- Store answers hashed/encrypted in database if feasible
- Rate limit repeated submissions (e.g., 5 attempts/minute)

17. Database Requirements

Tables / Collections

Teams

- Team ID
- Members
- Current Round
- Current Challenge
- Time
- Attempts

Challenges

- Challenge ID
- Answer
- Story Fragment
- Asset Paths
- Unlock Sequence

Logs

- Team
- Answer
- Time
- Result

18. Responsive Design

Desktop First

Should support

Laptop

Desktop

Tablet

Mobile compatibility desirable but puzzles are optimized for desktop.

19. Future Scalability

Architecture should allow

- Addition of new rounds
- Dynamic story fragments
- Multiple answer formats
- Hint system
- Team messaging
- Live announcements
- Real-time admin broadcasts

20. Recommended Tech Stack

Frontend

- Next.js (React)
- Tailwind CSS
- Framer Motion (animations)

Backend

- Node.js + Express
- Socket.IO (optional for live updates)

Database

- MongoDB (preferred for flexible challenge/story data)

Hosting

- Vercel (Frontend)
- Railway / Render / VPS (Backend)

Deliverables

The Tech Team should deliver:

1. Participant Authentication Module
2. Team Registration & Login
3. Progress Tracking System
4. Story Fragment Engine
5. Challenge & Answer Validation Engine
6. Embedded PDF/Image/Audio/Video Support
7. Terminal-Style UI Components
8. Leaderboard Module
9. Countdown Timer
10. Admin Dashboard with Qualification Override
11. Asset Management Panel
12. Submission Logging & Analytics
13. Secure Backend & Database Integration
14. Responsive, immersive spacecraft-themed UI